Whose Road Is Rough? Verified Trip Narratives from Vehicle Vibration and the Identifiability of Road Claims

Nishkarsh Singhal[1] and Geetanjali Bhola[1]

Faculty of Technology, University of Delhi, New Delhi, India

Abstract. A trip narrator built on vehicle vibration can check every sentence against its measurements and still be wrong about the road, because the vibration a stretch produces depends on the vehicle and on speed. We present RoadScribe, which turns IMU and GNSS recordings into narratives: it computes ISO 2631-1 vibration measures, segments the trip, compiles a ledger of typed facts, and lets a small language model write a narrative that a verifier checks claim by claim, falling back to a template narrator. We then ask when a road claim is identifiable from such data. On the PVS dataset, where three vehicles drove the same three routes, RoadScribe's per-stretch road class differed between two vehicles in 76% of comparisons, and the vehicle explained 18% of the variance of the road value (0.06% under a label-shuffling placebo). A speed elasticity estimated across places had the wrong sign (-0.64 versus $+0.86$ within places). [TBD: calibration result and SLM result.]

Keywords: Trip narration · Vehicle vibration · Identifiability · Data-to-text · Small language models · Road monitoring

1 Introduction

Vehicles and the phones in them can sense how a road shakes them, and road-monitoring work has used this for two decades to detect potholes, classify surfaces and estimate roughness [7,19,17]. Drivers, passengers and fleet operators ask questions in language rather than in roughness indices: which part of the route was rough, whether a jolt came from the road or from the driving, whether the map is wrong about a surface. Language models are a natural interface for such questions, and recent systems describe sensor data in text [31,16,33]. For a road description to be useful it must be true, and two kinds of truth are at stake. The first is *faithfulness to the measurements*: a narrative must not invent values or events. Data-to-text research has long addressed this with explicit content plans and claim checking [8,20,29]. The second is less often asked: *is the measured quantity a property of the road at all?* The vibration a stretch produces depends on the vehicle and on how fast it was driven. A narrator that says "this stretch is rough" on the basis of one vehicle's pass may be describing the vehicle.

This paper studies both questions with RoadScribe, a narrator for road trips recorded with an IMU and GNSS. RoadScribe computes ISO 2631-1 vibration

measures, segments the trip into road units, compiles a ledger of typed facts, and lets a small language model write a narrative that a verifier checks claim by claim against the ledger, with a template narrator as fallback. We then ask whether the road claims in its ledger are identifiable from a single trip.

Contributions.

- A verified narration pipeline (Sect. 3) and an honest benchmark of a fine-tuned 3B-parameter model against template narrators on nine real trips (Sect. 5.2). [TBD: one-line result of the regenerated S-PGR run.]
- A criterion for when a road claim is identifiable from a narrator's input, based on a standard source–channel model of vehicle vibration, and its consequences for single trips, fleets and speed (Sect. 4).
- A pre-registered cross-vehicle study on the PVS dataset [17] (Sect. 5.3–5.5). RoadScribe's road claims are not vehicle-invariant: the road class of a stretch differs between two vehicles in 76% of comparisons, and the vehicle explains 18% of the variance of the road value. A speed elasticity estimated across places has the wrong sign, and calibration on shared roads [TBD: P4 result].

2 Related Work

Road sensing from vehicles. Phones and in-vehicle units have been used to detect potholes and braking since Pothole Patrol and Nericell [7,19], and to classify surfaces or regress roughness from vibration, for example on the PVS dataset used here [17]. That the measured response depends on the vehicle and on speed is well established: road spectra and vehicle response are linked through the vehicle's transfer function [6,9], and crowdsourced methods estimate road roughness spectra from phone acceleration by inverting a vehicle model [3]. We do not propose a new roughness estimator. We ask which of these quantities a *narrator* may state as a fact about the road.

Language models over sensor data. LLMs have been prompted with raw sensor readings [31], aligned with motion-sensor encoders [16,10], given plots [32], trained on synthetic time-series question answering [30], and pre-trained on tens of millions of hours of wearable data paired with captions generated from statistics [33]. Their evaluation measures accuracy, retrieval or judged plausibility; the claims in their text are not checked against the measurements, and none of them asks whether a stated quantity can be determined from the input at all.

Faithful data-to-text. Separating content planning from realization is established in data-to-text generation [8,20], and so is measuring precision by extracting records from generated text and checking them against the source table [29]. Truth-conditional captioning checks time-series captions with executable programs [13], and factuality can be scored per atomic claim [18]. Sensor-to-text

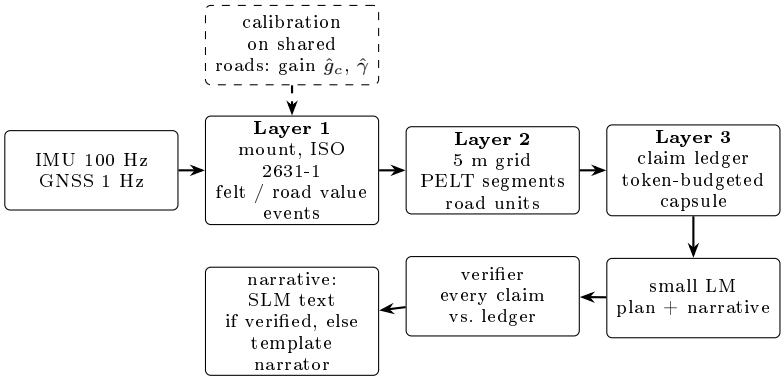


Fig. 1. RoadScribe. Values flow from the sensors through Layers 1–3 into the claim ledger; the language model writes a plan and a narrative over ledger facts, and a verifier either accepts the narrative or falls back to the template narrator. The calibration block (dashed) is the remedy studied in Sect. 4.

systems with audience-specific output go back to BabyTalk [22], and driver-feedback systems turn telematics into text with rules and, more recently, LLM-written tips [4,1,27]. RoadScribe's verifier and template fallback follow this tradition. Our contribution is the observation that verification against a ledger cannot catch claims whose ledger value is not identifiable, and a measurement of how often this happens for road claims.

3 RoadScribe

RoadScribe turns one recorded trip into a short narrative (Fig. 1). Three principles shape the design: every number and place in the text must trace to a fact in a ledger; the language model chooses and phrases content but never supplies a value; and when its output fails verification, a deterministic template narrator replaces it.

3.1 Layer 1: vibration measures

The IMU gives specific force $\mathbf{f}^b$ in the sensor frame at $f_s = 100$ Hz and GNSS gives speed at 1 Hz. The mount rotation $\mathbf{R}_{vb}$ is estimated without fixtures: the up axis is the geometric median of quasi-static gravity samples (Weiszfeld iteration with the Vardi–Zhang correction [28]), and the remaining yaw is the closed-form orthogonal Procrustes fit [24] of the levelled horizontal acceleration to GNSS longitudinal and centripetal acceleration. The gravity-free vehicle-frame acceleration is $\mathbf{a}^v = \mathbf{R}_{vb}\mathbf{f}^b - g\mathbf{e}_z$. ISO 2631-1 [11] weightings (W_k vertical, W_d horizontal) are applied exactly in the frequency domain, which is possible because narratives are written after the trip. For a stretch u the weighted RMS

values give the *felt* vibration $a_v = \sqrt{a_{wx}^2 + a_{wy}^2 + a_{wz}^2}$, and the original design defines a speed-decoupled *road* value

$$\tilde{a}_w = a_{wz}\,(v_{\mathrm{ref}}/\bar{v})^{\gamma}, \qquad v_{\mathrm{ref}} = 10\ \mathrm{m\,s^{-1}}, \tag{1}$$

reported only for $\bar{v} \geq 5\ \mathrm{m\,s^{-1}}$. The elasticity γ follows from road spectra of the form $G_d(n) \propto n^{-w}$ [12,6]: the RMS response of a fixed vehicle scales as $\bar{v}^{(w-1)/2}$, so $\gamma = (w-1)/2$. Both values are mapped to the six comfort classes of ISO 2631-1 Annex C with single cut-points $(0.315, 0.561, 0.894, 1.414, 2.236)\ \mathrm{m\,s^{-2}}$. Impulsive events are detected with the Teager–Kaiser energy operator [14] on the 4–25 Hz vertical band against a robust running threshold, and labelled with their maneuver context from GNSS and the gyroscope.

3.2 Layer 2: segmentation

Features are resampled on a 5 m along-track grid and the trip is segmented by penalized change-point detection [15], $\min_{\tau} \sum_i C(\mathbf{y}_{\tau_{i-1}+1:\tau_i}) + \sum_i \beta(\tau_i)$, with a Gaussian mean-shift cost. The design lowers the penalty $\beta(\tau)$ near map transitions. With a location-dependent penalty the textbook PELT pruning test is not exact; the correct test is $G(t) + C(\mathbf{y}_{t+1:s}) \geq G(s)$ with $G(t) = F(t) + \beta(t)$, a direct generalization of PELT's argument [15]. In 300 random series the textbook test returned a sub-optimal partition in 136, while the corrected test matched exhaustive optimal partitioning in all of them. Map matching is not implemented in the version evaluated here, so on PVS the penalty is constant and the segmentation is plain PELT. Short segments are merged, and each resulting road unit carries its Layer 1 measures and events.

3.3 Layer 3: ledger, capsule and generation

Units and events are compiled into a ledger of typed facts: severity classes (felt and road), measured values with tolerances, events with position and confidence, place anchors, comparisons, and *missing* facts that record why a value is unavailable (e.g. speed too low to separate road from speed). A capsule serializes the ledger within a token budget; units are chosen by an exact multiple-choice knapsack over salience. The small language model (Qwen2.5-3B-Instruct [23] with a QLoRA adapter [5]) receives the capsule and an audience (persona) and writes a plan over ledger identifiers followed by the narrative. A closed-vocabulary verifier extracts every claim from the narrative and checks it against the ledger: numbers within tolerance, severity words within their comfort band, events matched to ledger events, no value stated for a missing field. If the plan does not parse or any claim fails, the system returns the template narrator's text for the same capsule; this is a run-time-assurance pattern in the sense of the Simplex architecture [25]. The adapter was fine-tuned on 2,696 synthetic capsule–narrative pairs whose narratives come from the template narrator (2,500) and from a frontier model writing in the verifier's lexicon (196).

4 When Is a Road Claim Identifiable?

A verifier can only confirm that a sentence agrees with the ledger. Whether the ledger value is a property of the *road* depends on what the data can separate. For a linear vehicle with vertical transfer function H_c driving at speed v over a place p whose elevation spectrum is $\Phi_p(n)$ (n in cycles per metre), the vertical acceleration spectrum is [6,9]

$$S(f) = |H_c(f)|^2\, \Phi_p(f/v)/v, \qquad \log S(f) = A_c(f) + B_p(f/v) - \log v, \tag{2}$$

with $A_c = \log |H_c|^2$ and $B_p = \log \Phi_p$. Three consequences follow.

One vehicle, one pass. At a single speed, any function can be moved between A_c and B_p: the data fix only their sum. "This road is rough" and "this vehicle transmits a lot" are then indistinguishable. This is the indeterminacy known from separating spectra shifted by different velocities in astronomy [26,2]; we use it as a criterion for claims, not as a new result.

Several vehicles on the same places. If vehicles share places, the place term cancels in their differences, so the *relative* channels $A_c - A_{c'}$ are identified; the common part is identified (up to a constant) only if the same places are also driven at different speeds. On synthetic data with three channels and thirty road segments, a penalized Whittle fit [21] of Eq. (2) recovered the channel differences to within 0.02 log units in both cases, but the common channel only with speed diversity (error 0.010–0.023 versus 0.67–1.09 log units over three seeds).

Speed. For the band-integrated road value of Eq. (1),

$$\log a_{wz,cp} = \alpha_p + g_c + \gamma \log \bar{v}_{cp} + \varepsilon_{cp}, \tag{3}$$

with a place effect α_p and a vehicle gain g_c. The elasticity γ is identified by speed variation *within* places. Estimated across places within one trip, it also absorbs how drivers choose speed on different roads.

Criterion and remedy. We call a road claim *identifiable* from the narrator's input if its value is fixed by the data when the nuisances that can change (vehicle, speed) change. A narrator that states a non-identifiable value as fact is guessing even if every sentence passes the verifier, because the ledger stores the same confounded value. Equations (2)–(3) also give the remedy for a fleet: estimate $\hat{g}_c$ and $\hat{\gamma}$ on places that several vehicles have driven, with place and vehicle fixed effects, and report

$$\tilde{a}_{\mathrm{cal}} = a_{wz}\, (v_{\mathrm{ref}}/\bar{v})^{\hat{\gamma}}\, e^{-\hat{g}_c}. \tag{4}$$

This makes road claims comparable across calibrated vehicles. For a vehicle that has never shared a road with others, the road value remains unidentified, and the honest output is a statement about how the ride felt.

5 Experiments

5.1 Data and protocol

We use the Passive Vehicular Sensors (PVS) dataset [17]: nine trips in which three vehicle–driver pairs each drove the same three routes (10.7, 11.6 and

13.8 km). Each vehicle carries fixed IMU modules on the dashboard and above and below the suspension, a GNSS receiver at 1 Hz, and video from which the speed bumps, the surface type (asphalt, cobblestone, dirt) and the road quality of each side (good, regular, bad) were annotated. We use the left logger only (the two loggers have separate clocks that drift by up to 0.54 s) and its dashboard IMU, the closest proxy for a phone in a holder; the modules are not phones. To compare vehicles at the same places, each trip's GNSS fixes are matched in time order to the first vehicle's track within a forward window (the routes are loops, so nearest-point matching would confuse start and end). Matched fixes lie a median 1.5–3.8 m from the reference track, and 97.9–99.3% of samples are matched. Routes are cut into 100 m stretches, and a place enters the analysis if all three vehicles covered it at a mean speed of at least 5 $\mathrm{m\,s^{-1}}$, RoadScribe's own floor for road values: 310 places. The video labels agree across the three trips of a route (weighted $\kappa = 0.93$), so they serve as a vehicle-independent reference.

Hypotheses, thresholds and analyses for the cross-vehicle study were fixed and committed to the project repository before the vibration data were analysed; analyses added afterwards are marked as exploratory. Confidence intervals are from a block bootstrap over contiguous 1 km blocks of places (2,000 resamples), which respects spatial autocorrelation.

5.2 Narrator benchmark

[TBD: Table with T, T+V, S-PGR (regenerated with a sufficient token budget): acceptance, claims per narrative, P_L, H1 by cluster bootstrap over drives, grounding against labelled bumps with random-placement control.]

5.3 Are road claims vehicle-invariant?

Table 1 compares RoadScribe's two kinds of per-stretch claim across the three vehicles at the same places. Felt classes *should* differ, because comfort depends on the vehicle and the speed. Road classes are presented as properties of the road and should agree. They do not: the road class differs between two vehicles at the same place in 76% of pairs (by two or more classes in 14%), all three vehicles give the same road class at only 4.5% of places, and the weighted κ is 0.72 [0.60, 0.76]. In a regression of the log road value on place and vehicle fixed effects and log speed, the vehicle alone accounts for 18.2% [14.3, 27.5] of the variance. Shuffling the vehicle labels within places 1,000 times gives 0.06% on average (99th percentile 0.27%), so this vehicle signature is not an artefact of the analysis. The speed-decoupled road value is no more vehicle-invariant than the felt value (vehicle share 18.8%), and only 2.1% of the variance is a place-by-vehicle interaction: the vehicles differ mainly by a constant factor.

Sensor placement (exploratory). The size of the vehicle signature depends on where the sensor sits. For the module above the suspension it is small (vehicle share 2.9%, road $\kappa = 0.86$), and for the module below the suspension it is 11.8% ($\kappa = 0.60$). The dashboard, the placement closest to a phone in a holder, is the worst of the three.

Table 1. Agreement of per-stretch claims across the three vehicles at 310 places (dashboard IMU). κ: mean pairwise quadratic-weighted Cohen's kappa of the six comfort classes; vehicle share: unique variance explained by the vehicle in log value $\sim$ place + vehicle + log speed. Brackets: 95% block-bootstrap intervals.

Claim	κ	all three agree	pairs differ	vehicle share
Felt (a_v)	0.66 [0.53, 0.72]	9.7%	66%	18.8%
Road ($\tilde{a}_w$, Eq. 1, $\gamma = 0.35$)	0.72 [0.60, 0.76]	4.5%	76%	18.2% [14.3, 27.5]
Road, calibrated (Eq. 4)	[TBD]	[TBD]	[TBD]	[TBD]

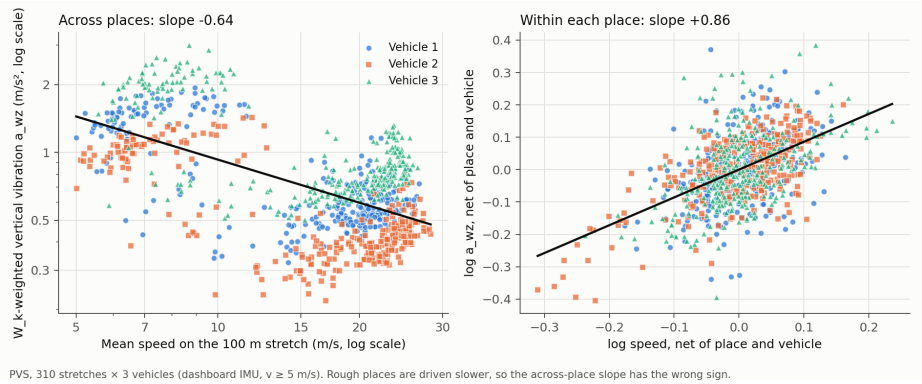


Fig. 2. Speed elasticity of dashboard vibration at 310 places, three vehicles. Left: across places (pooled OLS, slope -0.64). Right: within places, net of place and vehicle (partial-regression plot of Eq. (3), slope $+0.86$).

5.4 Speed

Fig. 2 plots the weighted vertical vibration of each place and vehicle against speed. Pooled across places, vibration *falls* with speed (slope -0.64): the slow places are the rough ones. Within a place, net of vehicle, it rises with slope $+0.86$ (the difference is 1.50 [1.26, 1.64]). The sign reversal comes mostly from surface type: cobblestone and dirt stretches are driven slowly, and within each surface the pooled slope is positive (asphalt $+0.16$, cobblestone $+0.54$, dirt $+1.36$; exploratory). Even within asphalt, however, the across-place estimate is about 80% smaller than the within-place one ($+0.16$ versus $+0.96$). The value $\gamma = 0.35$ in RoadScribe's default configuration was estimated in this across-place way, on asphalt with trip fixed effects (0.32 and 0.36 for the two dashboard bands); with it, the road value still rises with speed at a fixed place (slope 0.51). A speed elasticity for decoupling has to be estimated from repeated passes over the same places, as in Eq. (3).

5.5 Calibration on shared roads

[TBD: P4 results: leave-one-route-out calibration of $\hat{g}_c$ and $\hat{\gamma}$; agreement of calibrated road classes; template narratives before/after (contradictions per co-located stretch pair and number of claims); P2 gain-only vs spectral; P3 forward prediction.]

6 Discussion and Limitations

What verification does and does not guarantee. RoadScribe's verifier and template fallback guarantee that the narrative agrees with the ledger. The cross-vehicle study shows that this is a weaker property than it appears: the ledger's road values carry an 18% vehicle signature and a speed dependence that a single trip cannot remove, so a narrator can be perfectly faithful to its ledger and still contradict another vehicle's narrative about the same street. The remedy we found effective is simple and needs no new model: estimate a gain per vehicle and the speed elasticity on places that several vehicles have driven, and state road claims only for calibrated vehicles. A spectral factorization of the vehicle response [21] did not improve on this scalar calibration on PVS.

Limitations. PVS has three vehicle–driver pairs and one pass per vehicle and route, so vehicle and driver are confounded and measurement noise cannot be separated from the place-by-vehicle interaction. The sensors are fixed modules, not phones; the size of the vehicle signature depends on placement and may differ for phones in holders. Map matching is not implemented, so the map-informed part of the segmentation was not exercised. The language model was fine-tuned on template-generated narratives and therefore imitates the template; we did not evaluate question answering or audiences unseen in training, which require questions elicited from drivers and human judgements of audience fit. On-device latency and energy were not measured.

7 Conclusion

A narrator that checks every sentence against a ledger is only as trustworthy as the ledger is identifiable. On a public dataset with three vehicles driving the same roads, RoadScribe's road class for a stretch differed between two vehicles in 76% of comparisons, because the vibration a stretch produces depends on the vehicle and on speed in ways a single trip cannot separate. Calibrating each vehicle on shared roads, with the speed effect estimated within places, is the remedy, and narrators should state road properties only when such calibration exists. Code, pre-registrations and all analysis scripts are in the project repository.

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